

NxS

summary report for MN site

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1 Trial specs

1.1 fieldwork activities

4/14	Planted swath of experiment to MN Rothsay Spring Wheat. Truax planter and 5725? wheat tractor with gps. Seeded ~1.5 inches deep.
4/15	baseline soil samples collected by block. Starting plots on southernmost tire track. 4 pushes 0-6" for 0-6inch , 2 pushes for 6-24", by block.
4/15	Fertilized appropriate plots with pre weighed baggies of anvil treated urea and ammonium sulfate.
4/30	cultivated the northern and southern edge, potentially shrinking the harvest length of the northern and southern ranges/blocks
5/19	applied fertilizer at prescribed rates. Also flagged out field
5/23	ndvi measurements: 0.02 variance of ndvi in same plot. So plot 101 at 6877 could be remeasured at 7077 or 6677. Walked middle of the 8 ft plot, 15 ft length. Biomass sampling: 1 meter of row collected in middle of the plot at the base, placed in bag. Some plots done by kahlid were 3 rows at 1 meter instead of 1 row at 1 meter, we will figure out which plots by using weight of bags, then we will need to account for this at grain harvest as well.
5/23	sprayed 27 fl oz of wolverine on wheat (second application due to first one not being successful. Hoping for control of some grasses
6/24	Soil sampled every plot. 4 pokes 0-6 inches. 2 pokes 6-24.
7/24	staging check
7/28	started combining alleys. 1/4 completed, but settings mostly dialed in
7/30	combine harvest-
8/6	soil samples. 3 push 0-6", 1 push 6-24".

1.2 plot map

					north											
	NxS				west		east									
					south											
ft	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
25	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416
30	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316
30	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216
25	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116

Figure 1: NxS plot map. Plots were originally intended to be 30 ft long but the tractor + drill was not maneuverable enough to finish the 30 ft pass on the north and south side before needing to turn to avoid the hops. on most northern

2 Biomass

plant tissue analysis data still being processed

2.1 NDVI

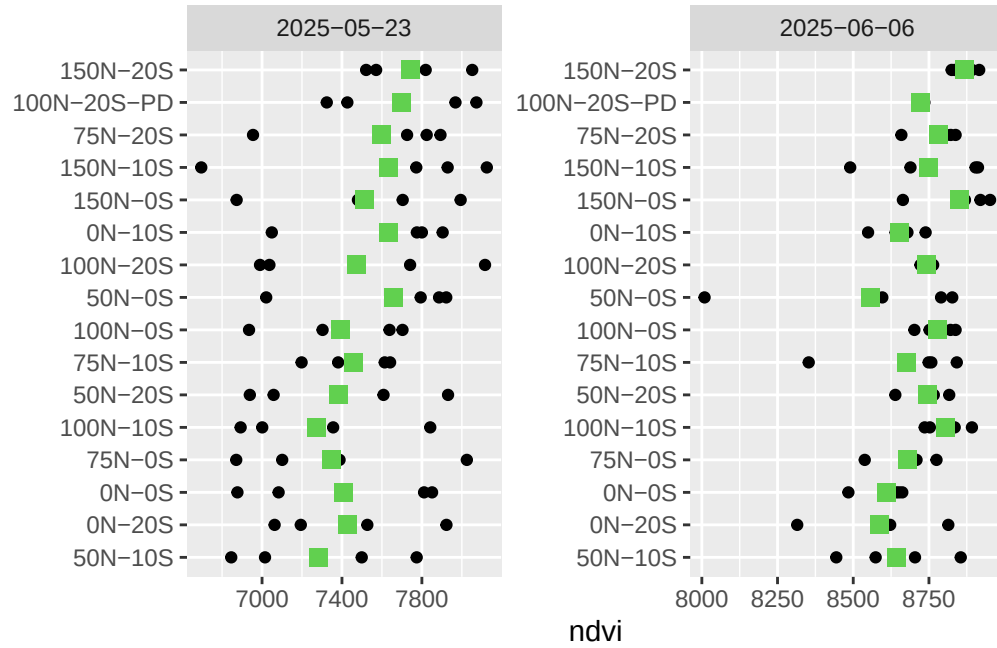


Figure 2: NDVI across two sampling dates. Green box indicates mean within sampling date. Treatments are ordered where 150N-20S is highest mean NDVI across both dates and 50N-10S is lowest mean NDVI when averaged across both dates.

2.2 Aboveground dry matter yield at feekes 3 (May 23)

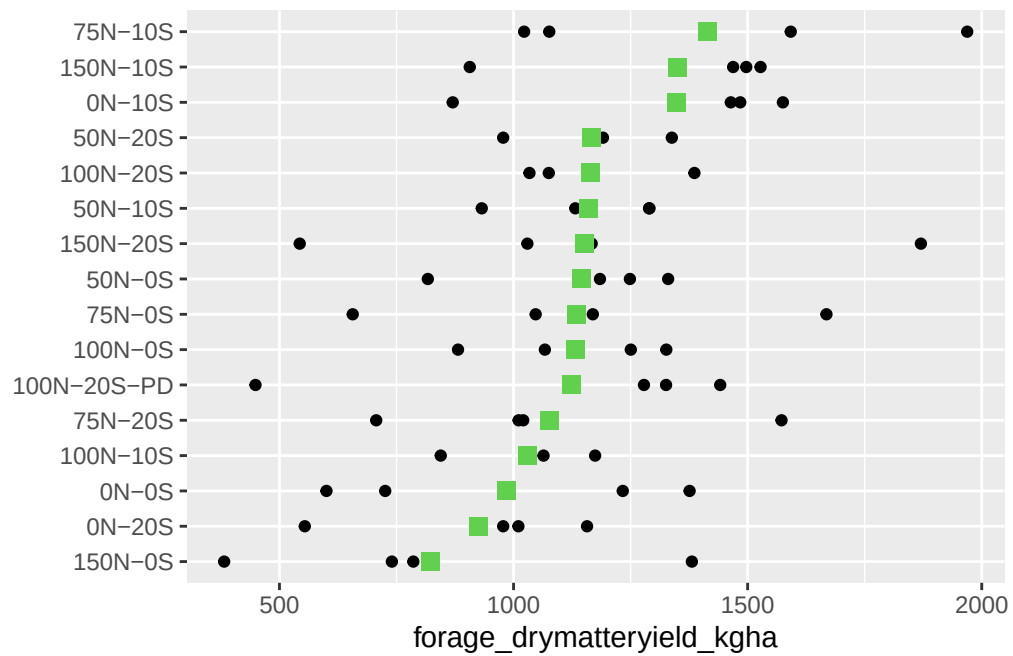


Figure 3: Forage dry matter yield on May 23. Green box indicates sample mean.

3 Grain (methods comparison)

Combine uses a harvest master system for plot yields, moisture and test weight

This was compared against a Perten 5200 moisture and test weight analyzer along with weighing the entire plot weight (bagged in the cab) on a scale after storing in a dryer for a couple days

3.1 kg ha

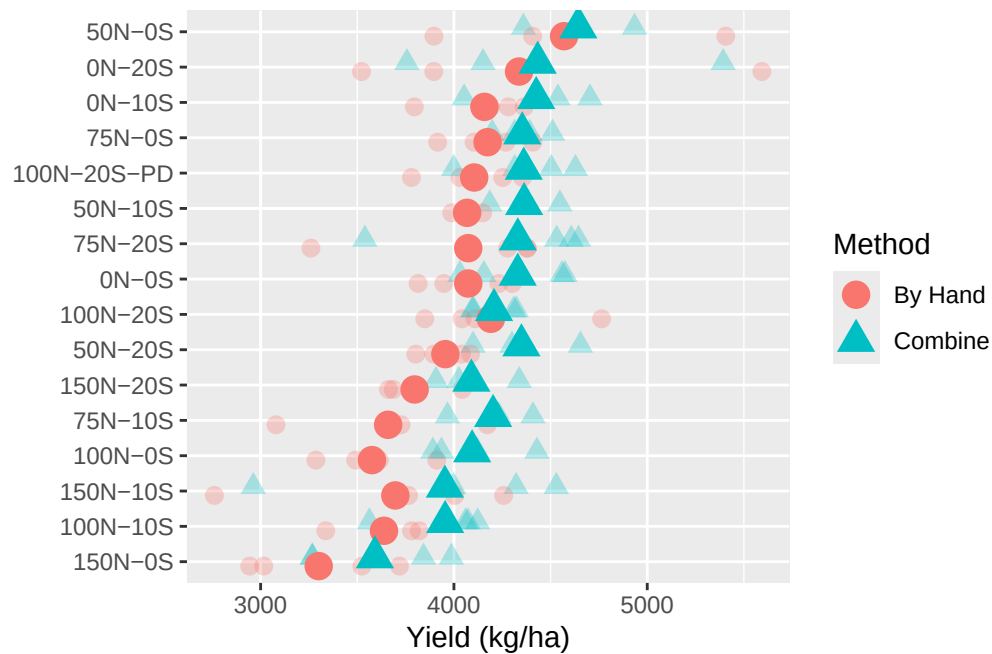


Figure 4: Grain yield in kilograms per hectare between the combine weight system vs the entire sample being brought back to lab and weighed. Note when weighed by hand the sample was dryer so this explains lower weights. Outlier removed.

- lower N treatments tended to yield higher. Lodging?

3.2 lodging

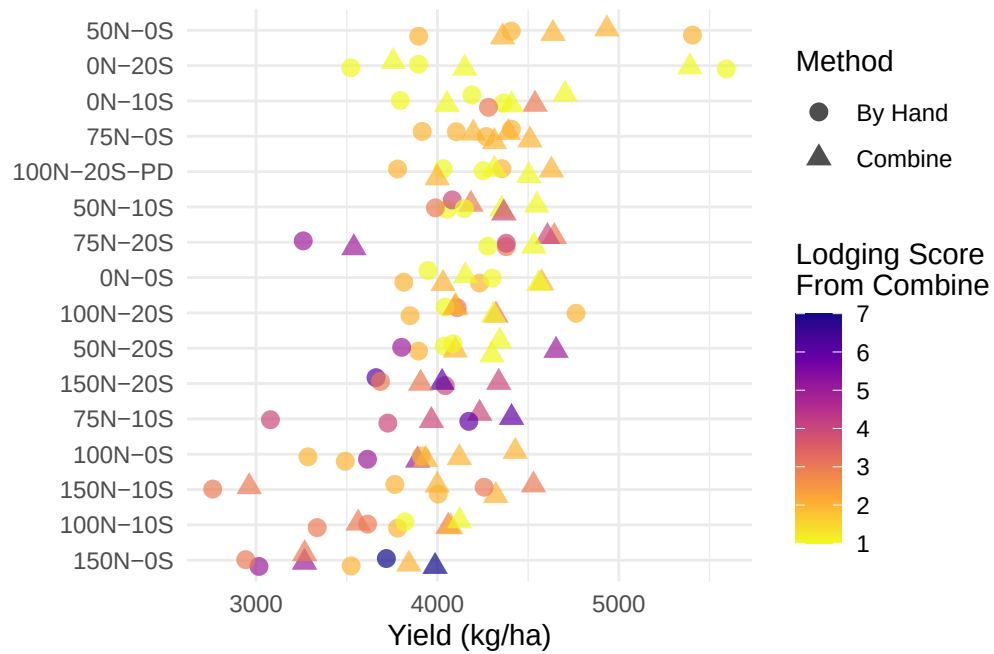


Figure 5: Grain yield estimation from combine with lodging scores where 1 = no lodging and 9 = flat on the ground.

- low yields driven by lodging
- I think drone scores were more accurate. See Figure 14.

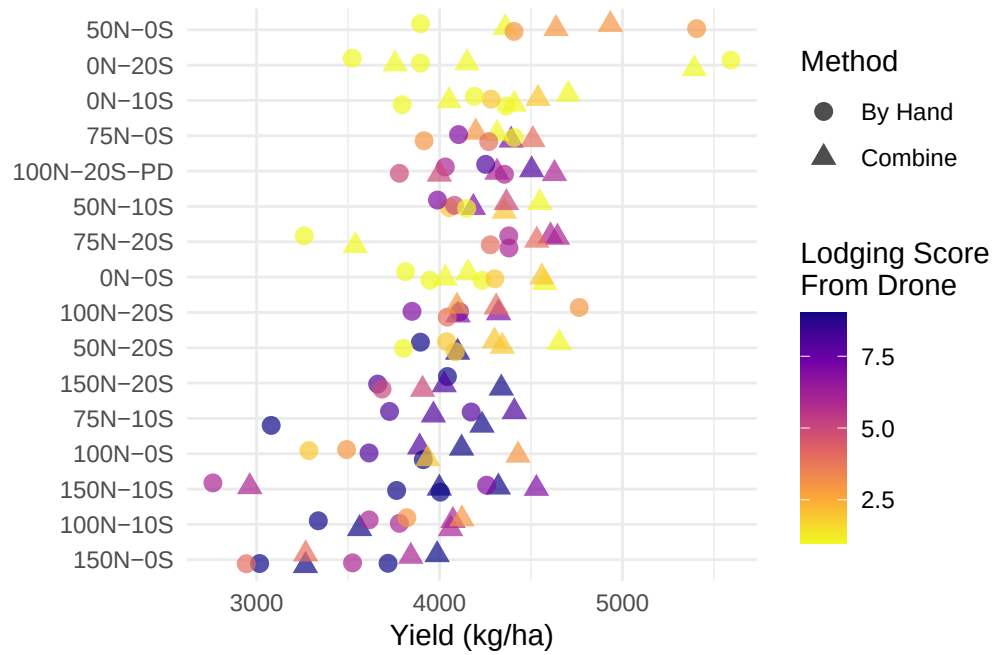


Figure 6: Grain yield estimation from drone photos with lodging scores where 1 = no lodging and 9 = flat on the ground.

3.3 test weight

- lower N treatments tended to have higher test weights.

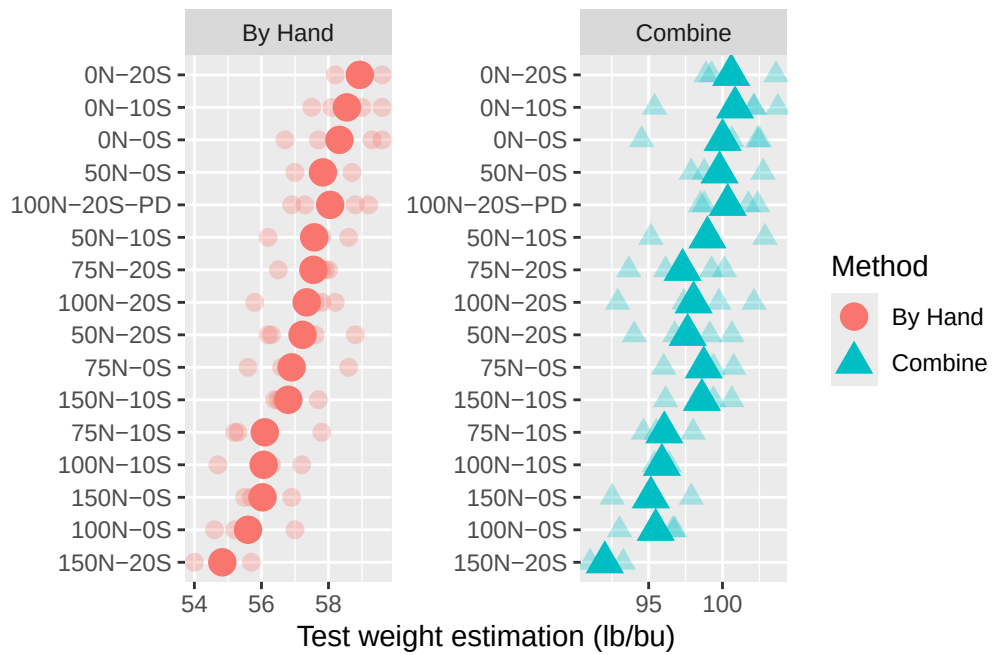


Figure 7: Grain test weight values were reasonable using using the Perten 5200 (i.e. By Hand) and were mostly mirrored by on board test weight estimator on combine, but those values will require a correction

3.4 bu acre

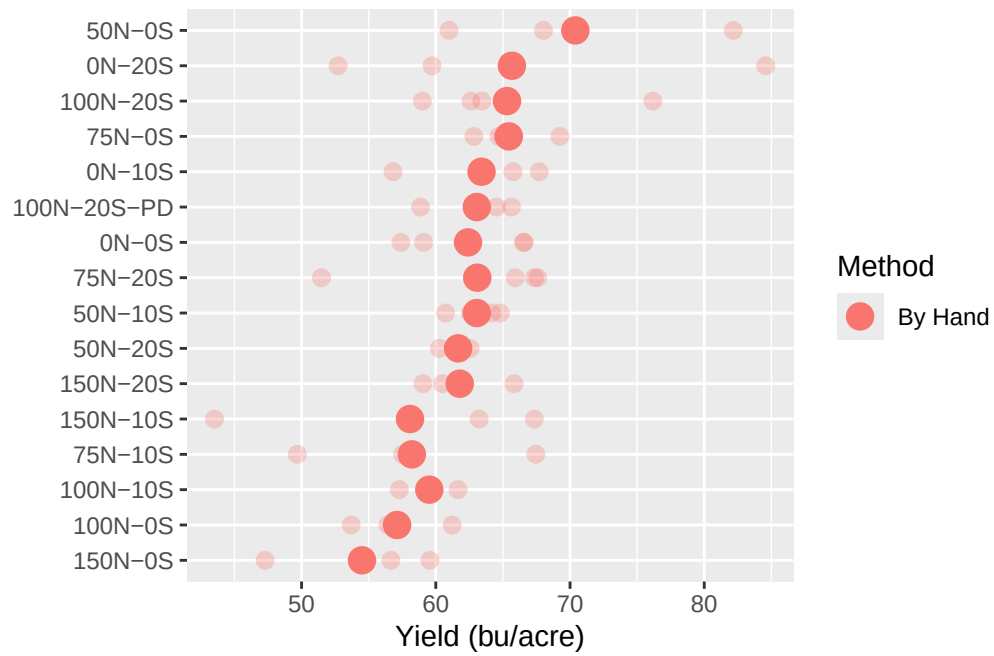


Figure 8: Grain yield in bushels per acre using only the by hand method because the test weight data was unreliable from the combine.

- high N treatments tended to do worse

3.5 moisture content

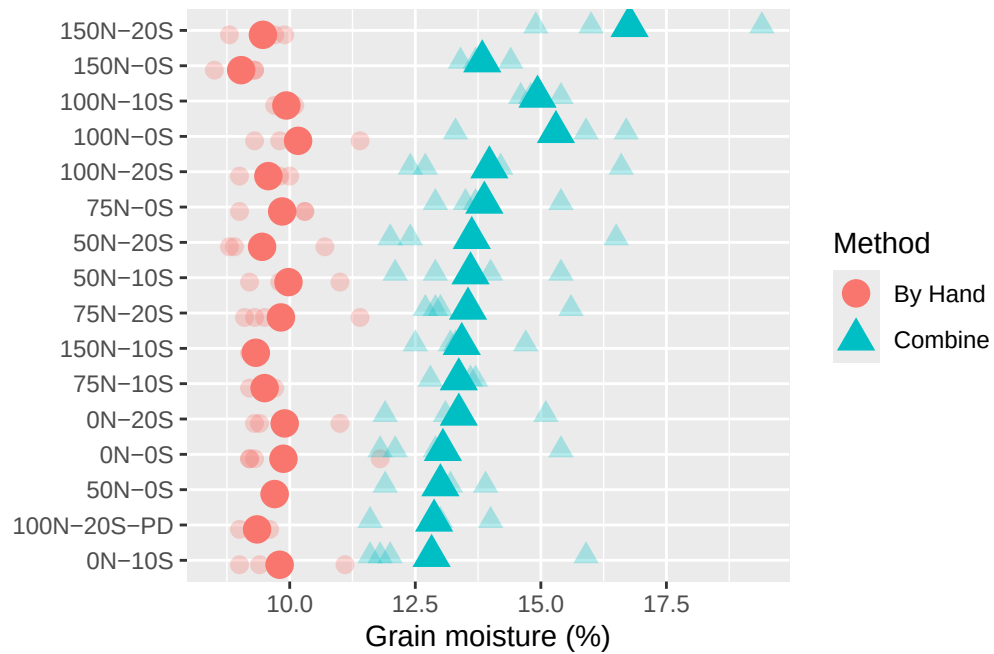


Figure 9: Moisture content at harvest (combine) and after dryer (By Hand).

- Higher N treatments had higher moisture at harvest
- All were dried down from 12-15% moisture at harvest to around 10% before the “by hand” weighing, test weight and moisture determination on the Perten 5200

4 Grain quality

4.1 protein

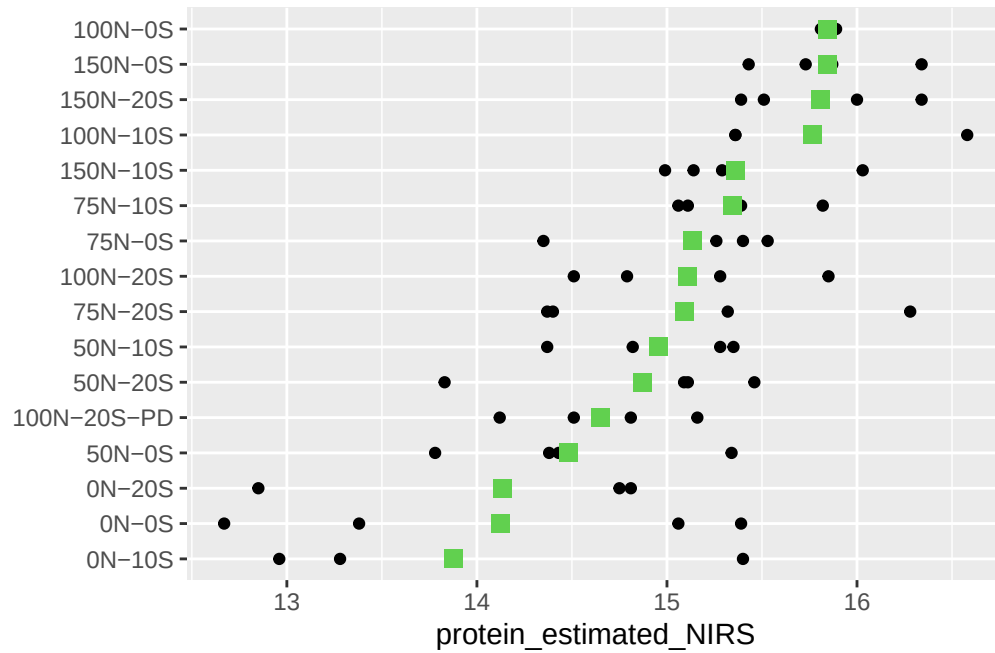


Figure 10: Protein content predicted by NIRS using wheat equation. Green box indicates mean.

- Higher N rates tended to have higher protein content

5 Soil

Not yet processed

6 Photos



Figure 11: Wheat stage on 21 May, 2 days prior to biomass sampling at Feekes 3 (though target was Feekes 5).



Figure 12: Drone photo from July 21 2025 prior to harvest. Note lodging.



Figure 13: Drone photo form July 21 2025 prior to harvest. plot borders and row+range ID's added.

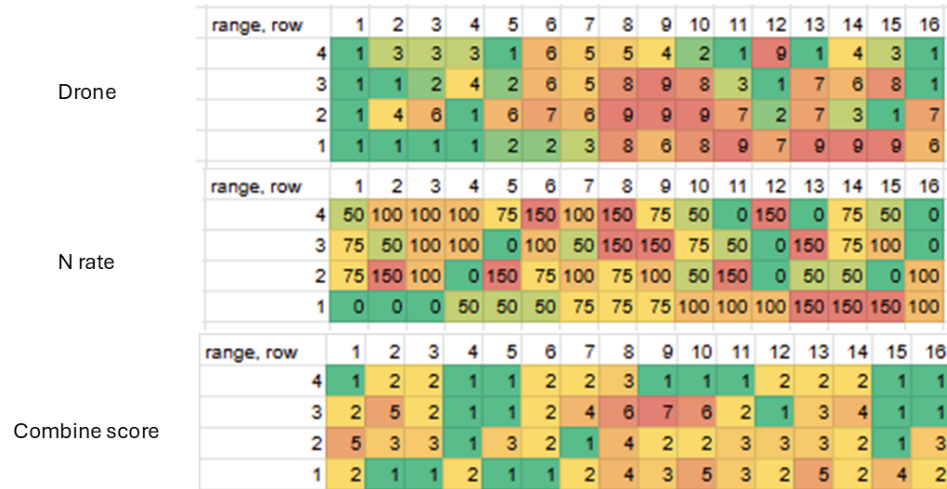


Figure 14: Heatmap of lodging scores taken either from the combine at harvest or using a drone image on 21July compared with the N rates of the treatments.

7 Discussion

7.1 Can we trust the combine?

7.1.1 Moisture

Probably. I think 13% moisture at harvest is reasonable, but didn't do a thorough comparison.

7.1.2 Test weight

No. Setting was set incorrectly. Combine trended similar to Perten 5200, but not exactly. That's why it was nice to have the backup of the benchtop Perten 5200

7.1.3 Plot yield

Probably. Didn't compare with quadrat harvest, but compared with taking the entire sample and weighing in the lab vs just taking a subsample we should be relatively confident we'll

capture treatment differences.

7.2 Lessons learned

Our lab is well suited to do trials like this. Plant all 1 variety. Multiple soil samples by plot. Tissue sampling. Combine harvest. We could easily become a site in other researchers trials and generate pretty good data and take good notes and photos and perform good QAQC on the data.

NDVI and biomass harvest data were too variable to be very useful in this trial.

Drone photos are incredibly valuable for verifying information like lodging and crop condition.